Quadratic Simultaneous Equations

Exam code: 9MA0

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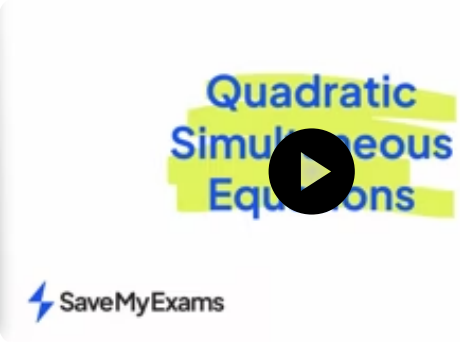
Updated on 19 June 2025



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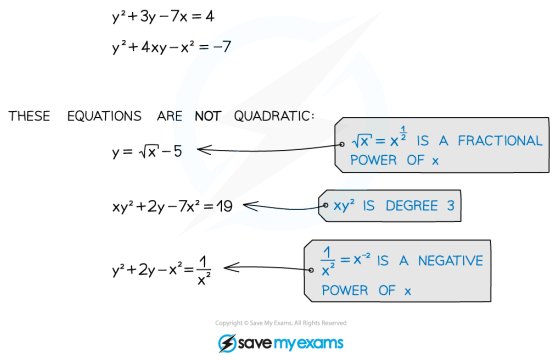
Quadratic simultaneous equations

What are quadratic simultaneous equations?

- When you have more than one equation in more than one unknown, then you are dealing with **simultaneous equations**
- An equation is quadratic if it contains terms of degree two, but no terms of any higher degrees (and also no unknowns raised to negative or fractional powers)

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



- Solving two simultaneous equations in two unknowns means finding pairs of values that make both of the equations true at the same time
  - At A level usually only one equation will be quadratic and the other will be linear
  - For one quadratic and one linear equation there will usually be **two** solution pairs (although there can be one, or none)

How do I solve quadratic simultaneous equations?

Step 1: Rearrange the **linear** equation so that one of the unknowns becomes the subject (if the linear equation is already in this form, you can skip to Step 2)

Step 2: Substitute the expression found in Step 1 into the quadratic equation

Step 3: Solve the new quadratic equation from Step 2 to find the values of the unknown (there will usually be two of these)

Step 4: Substitute the values from Step 3 into the rearranged equation from Step 1 to find the values of the other unknown

Step 5: Check your solutions by substituting the values for the two unknowns (one pair at a time!) into the original quadratic equation

Examiner Tips and Tricks

- You have to use substitution to solve quadratic simultaneous equations – the elimination method won't work.

Worked Example

$$2x - 3y = 23$$
$$3y^2 = 4x^2 + 11$$

$2x - 3y = 23$ ← LINEAR

$3y^2 = 4x^2 + 11$ ← QUADRATIC

STEP 1: $2x - 3y = 23$

$2x = 3y + 23$

← REARRANGE THE LINEAR EQUATION

BECAUSE THE QUADRATIC HAS $4x^2 = (2x)^2$ IT'S EASIEST TO STOP AT $2x$ HERE

STEP 2: $3y^2 = 4x^2 + 11$

$3y^2 = (2x)^2 + 11$

$3y^2 = (3y + 23)^2 + 11$

$3y^2 = 9y^2 + 138y + 529 + 11$

$6y^2 + 138y + 540 = 0$

$[\div 6]$ $y^2 + 23y + 90 = 0$

← SUBSTITUTE INTO THE QUADRATIC

STEP 3: $(y + 5)(y + 18) = 0$

$y = -5$

OR $y = -18$

← SOLVE BY FACTORISING (YOU COULD ALSO USE THE QUADRATIC FORMULA)

STEP 4: IF $y = -5$

$2x = 3(-5) + 23 = 8$

$x = 4$

IF $y = -18$

$2x = 3(-18) + 23 = -31$

$x = -15.5$

← SUBSTITUTE INTO REARRANGED LINEAR EQUATION

STEP 5: $3(-5)^2 = 4(4)^2 + 11$

$75 = 75$ ✓

$3(-18)^2 = 4(-15.5)^2 + 11$

$972 = 972$ ✓

← CHECK BY SUBSTITUTING INTO QUADRATIC!

THE SOLUTIONS ARE:

$x = 4$, $y = -5$

AND

$x = -15.5$, $y = -18$

← INDICATE WHICH x GOES WITH WHICH y

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